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Manifesting construction activity scenes via image captioning

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ABSTRACT

This study proposed an automated method for manifesting construction activity scenes by image captioning – an approach rooted in computer vision and natural language generation. A linguistic description schema for manifesting the scenes is developed initially and two unique dedicated image captioning datasets are created for method validation. A general model architecture of image captioning is then instituted by combining an encoder-decoder framework with deep neural networks, followed by three experimental tests involving the selection of model learning strategies and performance evaluation metrics. It is demonstrated the method's performance is comparable with that of state-of-the-art computer vision methods in general. The paper concludes with a discussion of the feasibility of the practical application of the proposed approach at the current technical level.

1. Introduction

Visual information relating to construction activity scenes are becoming increasingly important for construction management [1,2]. For instance, capturing the latest state of construction activities (especially direct work) by studying site images of scenes can be of significant help to project managers in meeting their objectives [2,4]. As scenes contain elements such as objects, relationships, and attributes [3], a construction activity scene in images can be defined as an integral whole view of the activity in pictures – synchronously containing objects (workers, equipment, materials, etc.), their interrelationships (e.g., cooperation between objects or coexistence of objects), and other essential scenario elements.

One current approach to capturing vision information from site images is by incorporating computer vision techniques with deep learning algorithms. This can be divided into four categories of object recognition [5–12], motion recognition [13–21], activity recognition [5,22–25], and scene analysis [26]. A limitation of these methods in the capture process and final manifestation is that the scene elements are captured separately (e.g., objects, object-object relationships, and objects with their attributes are treated independently). Another limitation is that, depending on the capture process, the final manifestation is typically generated as a predefined simple combination of semantic labels in the form of a single word or phrase. What is needed is an automated manifestation method that treats a construction activity scene as an integral whole by capturing the multiple objects, relationships, and related attributes from images synchronously and converting them into complete sentences.

Image captioning, which is rooted in computer vision (CV) and natural language generation (NLG), provides a solution by automatically representing scene information with both visual information and natural language sentences [27–32]. A completed image captioning method consists of a deep learning-based model and datasets. Recent advances recommend a general framework of image captioning model architecture that integrates a CNN “encoder” for extracting image information, word-embedding module, and an RNN “decoder” for generating image descriptions. However, existing benchmark image captioning datasets (e.g., Flickr8k [33], Flickr30k [34], and MS COCO [35]) involving general daily life cannot fit the unique scenario of construction activities – necessitating the creation of dedicated construction datasets. Then, the structural linkage between the visual information of scene elements and the corresponding natural language words needs to be identified.

In addressing these needs, this study involves two innovations. First, a novel linguistic description schema of a construction activity scene is proposed, which guides the development of the dedicated datasets. Second, a novel automated image captioning method is devised and validated for manifesting construction scenes. The image captioning model architecture of this method is established by integrating recent advances of image captioning in CV and NLG. Three experiments are

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described in which the datasets are used to train the image-captioning model, the results of which indicate the method's performance to be comparable with that of state-of-the-art computer vision methods in general. The paper then concludes with a discussion of the feasibility of the practical application of the method at the current technical level.

The remainder of this paper is organized as follows. Section 2 presents the background and related vision solutions for information manifestation in the construction domain and image captioning in CV and NLG; Section 3 outlines the method and materials used in developing the automated manifesting method; and the experimental testing is described in Section 4. Section 5 contains a discussion of the results of the study, its contribution to knowledge, practical implications, application challenges, and research limitations, while Section 6 provides some brief concluding remarks.

2. Background and related work

2.1. Vision solutions for manifesting construction activity scenes

Capturing vision information from site images of construction activities has generally involved the incorporation of adaptive computer vision techniques and deep learning algorithms. The differences between the different existing vision-based methods in the capture process and final manifestation can be divided into *object recognition*, *motion recognition*, *activity recognition*, and *scene analysis*.

Object recognition has been the subject of numerous studies in the construction domain over past decades. These commonly involve simple computer vision tasks (e.g., object detection, object tracking, and semantic segmentation). Deep learning-based methods are mainly used to detect onsite objects (e.g., sewer pipes [8]), monitor workers' unsafe behaviors (e.g., not wearing hardhats [7], not wearing safety belts [6], and misusing ladders [5]), and track the location of construction equipment (e.g., backhoes, automobiles, trucks, compactors, excavators, and workers [11,12]). Semantic segmentation by color-based labeling has been used to identify and list whole objects in an image [9,10]. For example, Kim et al. [9] present a data-driven scene parsing method to recognize various objects in a construction site image. Rahimian et al. [10] develop a system that includes object recognition and semantic segmentation to identify different structural elements from a visual representation for superimposing real-world images on an ‘as-planned’ model. All these methods involving object recognition are implemented using a preexisting structure between labels to train better models [36]. Such fixed semantic relationships of vision are only suitable for object recognition tasks and cannot represent typical complicated construction activity scenes with diverse combinations of objects and relationships.

Motion recognition in the construction domain uses a systematic approach to understand and analyze the camera-captured movement of construction workers and equipment [37]. This mostly involves posture estimation of *workers* and *equipment*. *Worker posture recognition* is mainly applied to estimating the workers' state of health and safety behaviors. On the one hand, methods have been developed for the workload estimation and physical fatigue of construction workers [14–17]; for example, Yu et al.'s [14] automatic workload estimation using image-based three-dimensional (3D) posture-capture smart insoles to assess body joints, and a biomechanical analysis to provide accurate assessments based on real data instead of simulation. On the other hand, to monitor and assess ergonomic hazards on site, Yan et al. [19] address construction hazard prevention by developing an ergonomic posture recognition technique from view-invariant features in 2D skeleton motion based on an ordinary 2D camera. In contrast, *equipment posture recognition* has mainly involved recognizing single actions of earthmoving construction equipment based on computer vision algorithms [13]. Automatic equipment posture estimation focusing on the action of equipment has been implemented to improve the recognition efficiency of equipment work states [20] and enhance equipment productivity (e.g., excavators) [18]. A further study has even developed and demonstrated a novel method for automatically estimating the full-body posture and movement of onsite construction equipment that can avoid potential collisions and other accidents for safer onsite conditions [21]. These vision solutions involving motion recognition are based on ergonomics: integrating object recognition and pattern recognition, with the motion information being obtained by detecting the key points of the object's structure (e.g., the joint of a human body or angle of an excavator's arm) and analyzing the inner time-spatial interrelationships involved.

Activity recognition, defined as matching the relationship between detected objects with an activity pattern label, has emerged recently – mainly involving direct work with the worker's action [5,22–25] – with these relationships being represented as the cooperation between objects or coexistence of objects [24]. Yang et al.'s [22] comprehensive study of worker actions abstracted 11 common types of actions from 5 different trades, and established a new real-world video dataset with 1176 instances. A cutting-edge video description method with dense trajectories has also been applied for action recognition. Considering the challenges in a specific application scenario, such as safety, Ding et al. [5] develop a new hybrid deep learning model by integrating a convolution neural network (CNN) and long short-term memory (LSTM) that automatically recognizes the unsafe actions of workers. Luo et al. [24], on the other hand, have introduced a two-step method for recognizing diverse construction activities in still images. In doing this, 22 classes of construction-related objects using convolutional neural networks were detected initially, then they defined semantic relevance (representing the likelihood of the cooperation or coexistence between two objects in a construction activity), spatial relevance (representing the two-dimensional pixel proximity in the image coordinates), and activity patterns to recognize 17 types of construction activities. After further improvement, construction activity recognition was used to measure the worker productivity of the site [23]. Although these activity recognition methods focus on exploring the relationships between objects, which are more complicated than simple object recognition tasks, the final manifestations are limited to representing an action or type of work state.

Scene analysis has received relatively little attention in the construction domain to date. It has been used in semantic segmentation to detect and label all objects in a whole picture [9,10] and to classify site pictures by matching the scenes involved with specific safety rules [26]. Specifically, a crowdsourcing approach for the safety inspection is examined, an approach developed to label images of complex construction scenes having safety-rule violations, with a focus on minimizing false alarms while keeping within acceptable rates of false negatives. For information capturing and manifestation, this approach considers the construction scene as an integral whole for recognition and matching the semantic relationships between the scene's vision features and manual text labels representing the safety rules. The final manifestations are the combination of the scene images and text of safety rule violation labels. The safety rule violation labels manifested only correspond with the captured matching relationships but ignore the related inner elements.

Fig. 1 summarizes the existing vision solutions in depicting some sample images of the four categories of information manifestation from ten published papers in the construction domain, and Table 1 presents the process captures and final manifestations – the images in Fig. 1 corresponding with each category of existing methods in Table 1. This shows that the scene elements are captured separately: the objects, object-object relationships, and objects with their attributes are treated independently. In addition, depending on the capture process, the final manifestation is typically generated as a predefined simple combination of semantic labels in the form of a single word or phrase (e.g., objects are labeled with nouns, and relationships are labeled with verbs or verbal phrases). Taken alone, therefore, current vision solutions are inappropriate for manifesting construction activity scenes as an integral

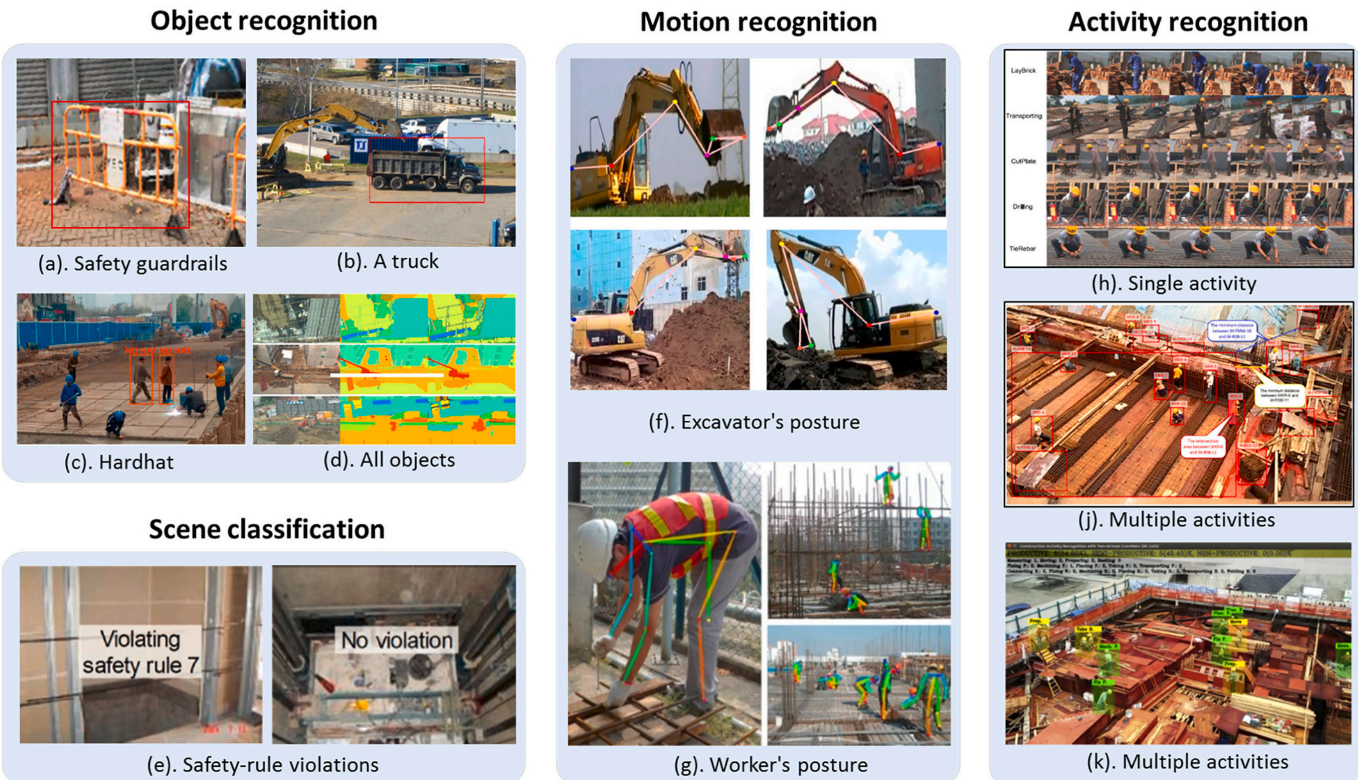
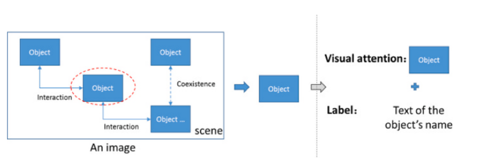
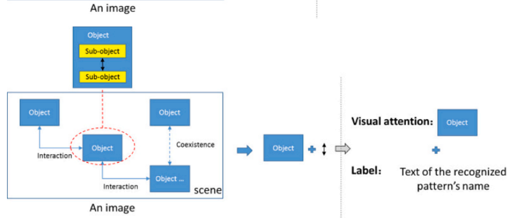
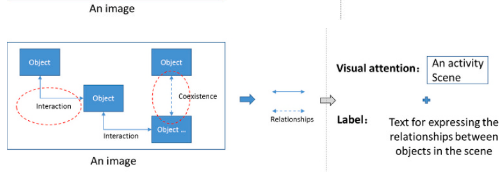
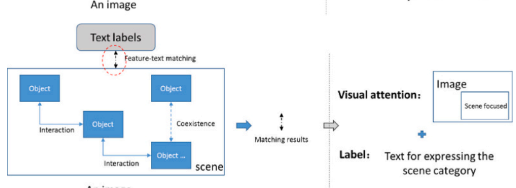


Fig. 1. Sample images from previous studies of vision-based methods to manifest information in the construction domain, comprising (a) safety guardrails detection [38], (b) equipment detection, (c) worker hardhat detection [7], (d) object recognition in whole images with semantic segmentation [9], (e) scene analysis for recognizing safety-rule violations [26], (f) excavator posture estimation [21], (g) worker posture estimation for avoiding hazards [19], (h) single activity recognition [22], (j) multiple activity recognition on site [24], and (k) multiple activity recognition for work sampling [23].

Table 1
Captures and manifestations by using existing vision-based methods in the construction domain.

Methods	Schematic of the process	Captures	Final manifestation	Samples in Fig. 1
Object recognition		An object	• *Visual attention*: object • *Labels*: text of the object's name	(a); (b); (c); (d)
Motion recognition		A pattern relationship within an object	• *Visual attention*: object with its motion • *Labels*: text of the recognized pattern's name	(f); (g)
Activity recognition		A relationship (cooperation or coexistence) between objects	• *Visual attention*: an activity scene with related objects • *Labels*: text for expressing the relationships between objects in the scene	(h); (j); (k)
Scene analysis		A matching relationship between scene focus and predefined text labels	• *Visual attention*: image with the focused scene • *Labels*: text for expressing the scene category	(e)

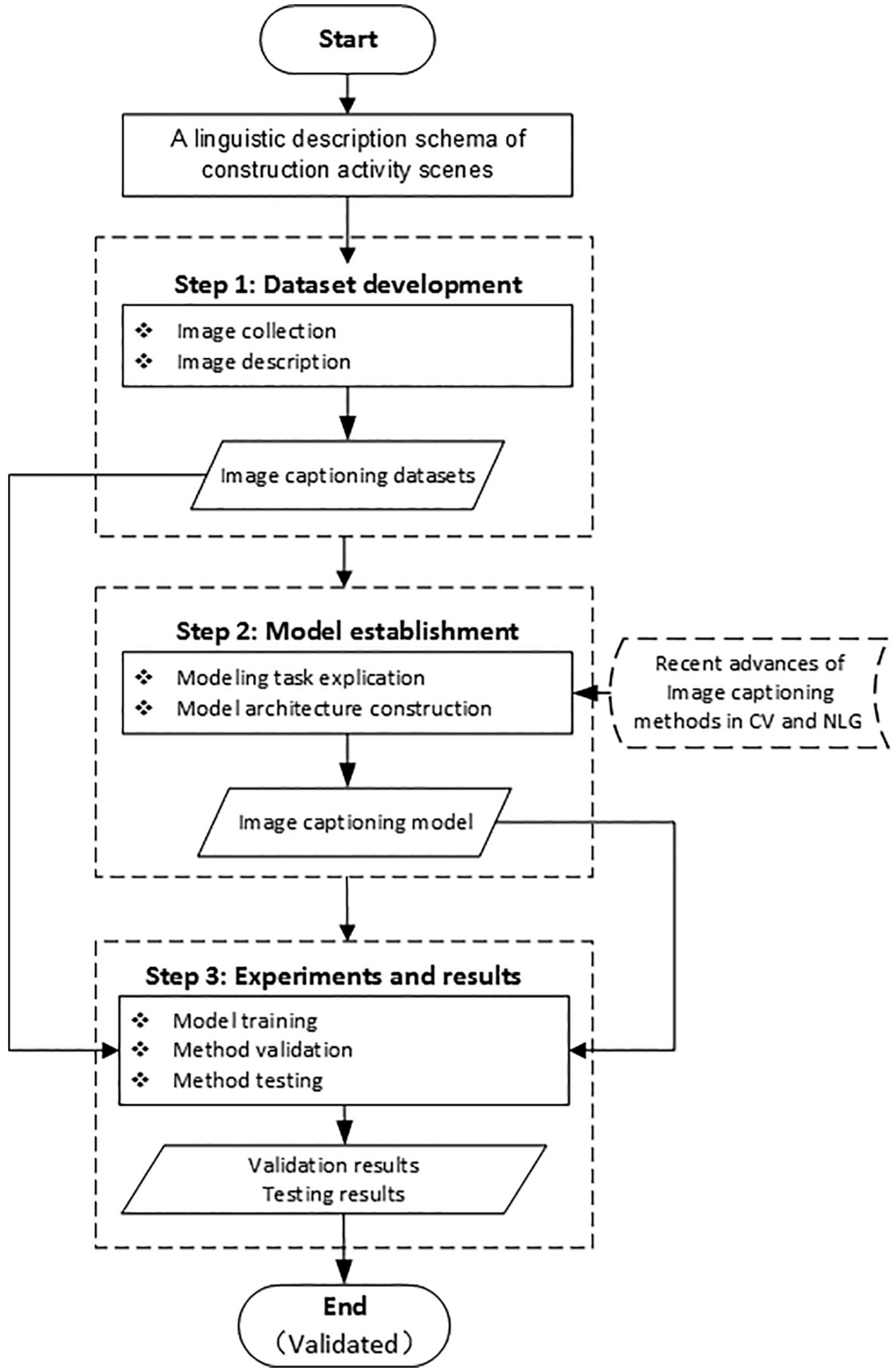


Fig. 2. Steps in method development and testing.

whole, which needs to capture multiple objects, relationships, and related attributes synchronously. A potential solution is image captioning, which can be used to manifest the construction activity scene with both visual information and natural language descriptions [36], as described in the next section.

2.2. Image captioning

Several studies of automated image captioning methods in the fields of CV and NLG have emerged in recent decades, but as yet with no related construction management applications. This section presents recent advances in image captioning to identify the general model architecture and the gap to be bridged between the existing methods and scenario applications.

As a branch of the natural language processing (NLP) field, NLG is defined as "the subfield of artificial intelligence and computational linguistics that is concerned with the construction of computer systems that can produce understandable texts in English or other human languages from some underlying non-linguistic representation of information" [39]. While NLG can involve either text-to-text or data-to-text considerations, image captioning is arguably a paradigm case of data-to-text generation where the input is in the form of an image [40]. Recent studies show that such NLG architectures and methods as deep neural networks (DNNs) and the encoder-decoder framework are now

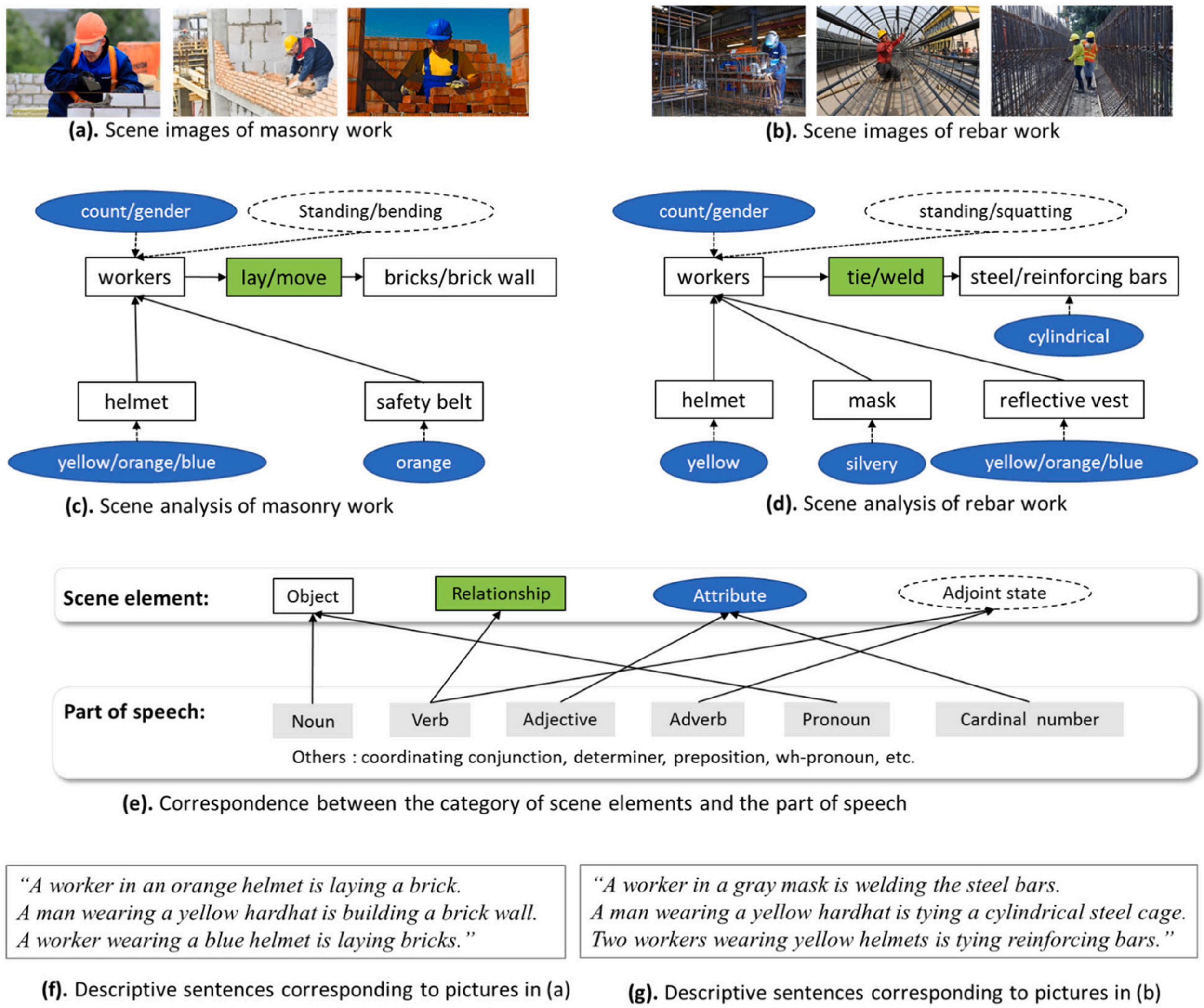


Fig. 3. Sample linguistic description schema of construction activity scenes.

also the dominant approach in image captioning [40]. *DNNs* are designed to learn representations at increasing levels of abstraction by exploiting backpropagation [41] and have had notable successes in sequential modeling using feedforward networks [42,43], log-bilinear models [44], and recurrent neural networks (RNNs) [45], including RNNs with long short-term memory (LSTM) units [46] – now the dominant type of RNN for language modeling tasks. The *encoder-decoder framework* is an influential architecture for NLG [47], where an RNN is used to encode the input into a vector representation, which is also the auxiliary input of the decoder RNN. This decoupling between coding and decoding makes it possible to share coding vectors across multiple natural language processing tasks in a multitasking learning environment [48]. The encoder-decoder framework is well-suited to such Sequence-to-Sequence (seq2seq) tasks as machine translation, where the variable-length input sequence in the source language needs to be mapped to the variable-length sequence in the target [49], making it easy to apply the framework to NLG: for example, seq2seq models have also been used to generate text from abstract meaning representations [50].

The traditional approach to establishing image captioning models followed machine translation [51], after which a method was developed that can express the possibility of linking an image to a sentence by computing a score [52]. Recently, it has been shown that convolutional neural networks (CNNs) can produce a rich representation of the input image by embedding it in a fixed-length vector such that this representation can be used for a variety of vision methods [53]. Many methods inspired by the encoder-decoder framework in machine translation (details in Section 3.3) have been proposed for image captioning, based on a CNN as the encoder and an RNN as the decoder [27,28,30,32,54–57] – the LSTM model being one of RNNs predominantly used for sentence generation [27,30,54,55,57]. Moreover, an alignment model has been proposed based on a structured objective aimed at two modalities through a multi-model embedding, a novel combination of CNNs for representing images, and bidirectional RNNs for disposing of sentences [32]. Subsequently, the results of the image captioning model based on the above core concept (of combining a CNN encoder with an RNN decoder) were further improved by using the LSTM model as the RNN decoder for sentence generation [30].

Datasets are also vital for image captioning methods due to them being on deep learning models; big data for training such models is essential. Several benchmark image captioning datasets exist (e.g., Flickr8k [33], Flickr30k [34], and MS COCO [35]), containing tens and hundreds of thousands of images involving general daily life, but none involve the complicated construction site environment.

A general image captioning model architecture can therefore be constructed using an "encoder" CNN to extract the features of images, and a "decoder" RNN to generate sentences by connecting embedding words with an image semantic sequence. Hence, the remaining work in establishing a model mainly involves selecting the latest DNNs (an "encoder" CNN and a "decoder" RNN) and adopting adaptive training

Table 2
Overview of the scene image-sentence pairs for the five categories of construction activities.

Scenes	Scene elements		
	Main objects	Main actions	Main attributes
Cart transportation	Worker; cart; safety measurements	Pull; push; unload	Color; count; gesture
Masonry work	Worker; bricks; mortar; safety measurements	Lay; fix; cut; measure; build; move	Color; count; gesture
Rebar work	Worker; reinforcing/steel bars; safety measurements	Tie; shear; weld	Color; count; gesture; shape
Plastering	Worker; mortar; safety measurements	Plaster; mix	Color; count; gesture; shape
Tiling	Worker; mortar; hammer; safety measurements	Lay; stick; fix; cut	Color; count; gesture; shape

strategies and evaluation metrics. However, existing image captioning methods, involving both the model architecture and general-life datasets, are unsuited to manifesting construction activity scenes because of the lack of a structural linguistic description schema of construction activity scenes and the datasets for method validation. What is needed is a merging of the state-of-art image captioning techniques and linguistic description analysis.

3. Method

Fig. 2 shows the steps taken in developing a scenario-based image captioning method for construction activity scenes, including dataset development, model establishment, and experimental testing. The approach described here involves the development of 1) a linguistic description schema of construction activity scenes, 2) the datasets, and 3) the image captioning model.

3.1. Linguistic description schema of construction activity scenes

Confirming a structural linguistic description schema of construction activity scenes is the primary consideration in developing an automated method via image captioning for their manifestation. This involves 1) deconstructing the scene image by transferring the activity-related vision information into scene elements based on the concept of the scene, 2) selecting the words with the correct parts of speech to match the identified scene elements, and 3) forming logically and correct descriptive sentences by adding the necessary conjunctions or transformations.

Fig. 3 provides a sample linguistic schema of masonry work and rebar work scenes. Fig. 3(a) and (b) contain a of pictures of masonry and rebar work scenes respectively; Fig. 3(c) and (d) contain scene elements corresponding to the images in Fig. 3(a) and (b) respectively, connected by lines and arrows based on their real-world relationships. Here, the white boxes represent the main objects to be identified in the picture, the green boxes represent the actions describing the relationship between the recognized objects, and the blue ellipsoid boxes represent such basic attributes of objects as color, shape, size, and count. There is an additional attribute for the "workers" representing their adjoint behaviors, e.g., gestures of the worker. A complete scene of site photo should therefore contain multiple scene elements (objects, actions, and attributes) simultaneously. Fig. 3(e) illustrates the correspondence between the scene elements and the parts of speech. The corresponding manifestations with structural descriptive sentences are presented in Fig. 3(f) and (g).

3.2. Image captioning datasets

Two image captioning datasets of different sizes and different categories of construction activity are developed. The next two sections describe the guidelines used in developing the datasets and provide an overview of the resulting datasets.

3.2.1. Guidelines for developing the image captioning datasets

1) *Image collection.* Images with construction activity scenes can be collected from real-world construction sites using digital cameras. These need to meet two requirements: (1) each image should focus on at least one construction activity; and (2) the number of images is enough for deep learning.
2) *Image description.* In one scene image, the interactions between objects are highly complex, going beyond simple pairwise relations [3], so multiple sentences with different combinations of scene elements (objects, relationships, and attributes) are used to describe the scene. Guided by the adopted manifestation schema in Fig. 3, one logic sentence should reflect one complete construction activity, including objects, relationships, and attributes. Each image is described with five sentences, following the rules of creating the benchmark datasets (e.g., Flickr8k [33], Flickr30k [34], and MS COCO [35]). The process of image description is carried out manually. In addition to the description of the object having to be related to construction activity scenes, the syntactic requirements of other aspects follow the production instructions of the MS COCO caption dataset [58].

3.2.2. Developing process of two datasets

Based on the above general principals and instructions, two image captioning datasets are created of different sizes and with different construction activity categories in the following steps.

1) *Selection of construction activity categories.* Five categories of construction activity scenes are chosen: cart transporting, masonry work, rebar work, plastering, and tiling. All the photographs are taken on actual construction sites. Table 2 summarizes the scene elements in each category.
2) *Classification of scene elements.* Following the different construction management roles involved, the main scene elements are classified into the five categories (see Fig. 4) of: 1) *subjects* (i.e., "workers") that directly involve site safety, expressed as Object I; 2) *objects* (e.g., "carts", "steel/reinforcing bars", "bricks", "wall", and "ceramic tile") that directly represent tools or materials of the activity, expressed as Object II; 3) *safety apparel* (e.g., "helmets", "reflective vests", and "safety belts") that are always attached to the first class of objects, expressed as Object III; 4) *object relationships* between Object I and Object II, with different verbs being used to describe the interactions involved; and 5) *object attributes*, mostly concerning "color" and "count".
3) *Image description.* The descriptive sentences of images are organized manually in complying with the MS COCO caption dataset production instructions [58]. The substitution of synonyms, conversion of active and passive voices, addition and deletion of accompanying features and attributes, and other forms of sentence transformation are also considered in addition to emphasizing objectivity rules. The synonyms or word neighbors are summarized and classified in advance (see Table 3) to avoid biasing the statistical results. Fig. 5 shows the ultimate captioning examples of the five activity types.
4) *Data preprocessing.* All the selected images and corresponding

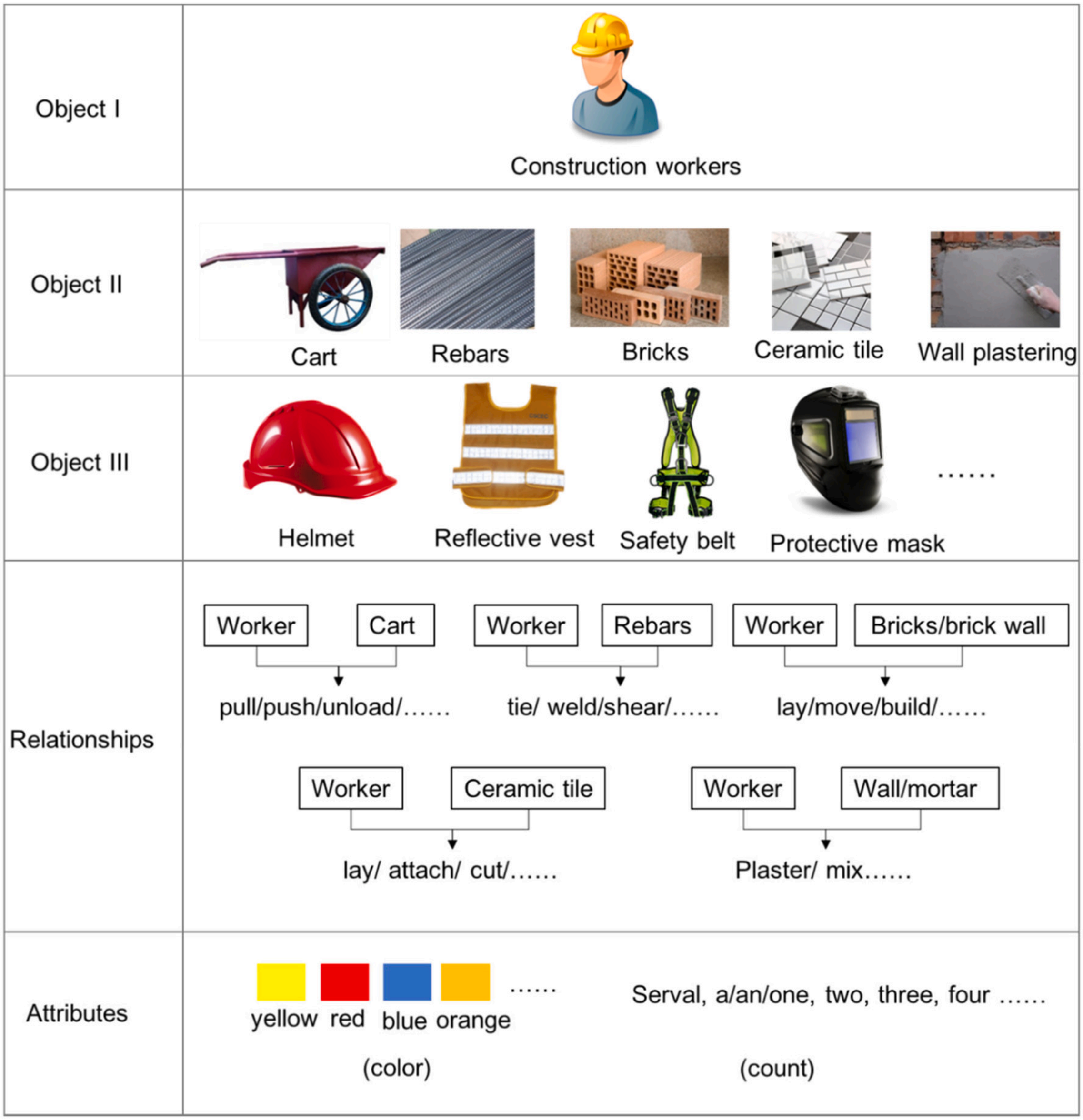


Fig. 4. Classification of the main scene elements.

Table 3
Some example words with their synonym or nearest neighbors.

Word	Synonym/Neighbors
Worker	Man/men, woman, construction worker
Cart	Van
Rebar	Steel-bars, reinforcing-bars
Brick	Block
Helmet	Hat, sunbonnet, hardhat
Reflective vest	Safety belt

descriptive sentences are pairwise coded for model learning. Before training, all the images are scaled to 224 × 224 pixels, and captions mapped to images in the form of JSON files.

Table 4 presents an overview of the two datasets. Dataset I comprises 2400 (i.e., 480 × 5) image-sentence pairs for three categories of activities (i.e., cart transportation, masonry work, and rebar work). Dataset II comprises 34,510 (i.e., 6902 × 5) image-sentence pairs for all categories of activities in Table 2. In each dataset, 66% of the image-sentence pairs are used as the training set, 17% form the validation set, and the other 17% form the testing set. The part-of-speech distributions of Dataset I and Dataset II are compared by using the NLTK toolkit [59] in Fig. 6, as the part-of-speech can partially reflect its role in construction scenes; for example, "NN" and "NNS" represent the nouns that describe the still-objects onsite photographs, and "VB" represents the verbs that describe the relationships between different objects. The words that represent the scene elements are termed "scene words".

3.3. Image captioning model

The image captioning model is established by two steps: 1) explicating the modeling task by reviewing recent advances in image captioning, and 2) constructing its general architecture.

3.3.1. Modeling task explication

The fundamental theoretical framework for image captioning is mainly inspired by the machine translation model structure [27,30], as Fig. 7 shows. An image I (instead of an input sentence in a source language) is "translated" into its description D, with the RNN "encoder" replaced by a CNN, which has been widely used to study image tasks and is the state-of-the-art in object recognition and detection. In this

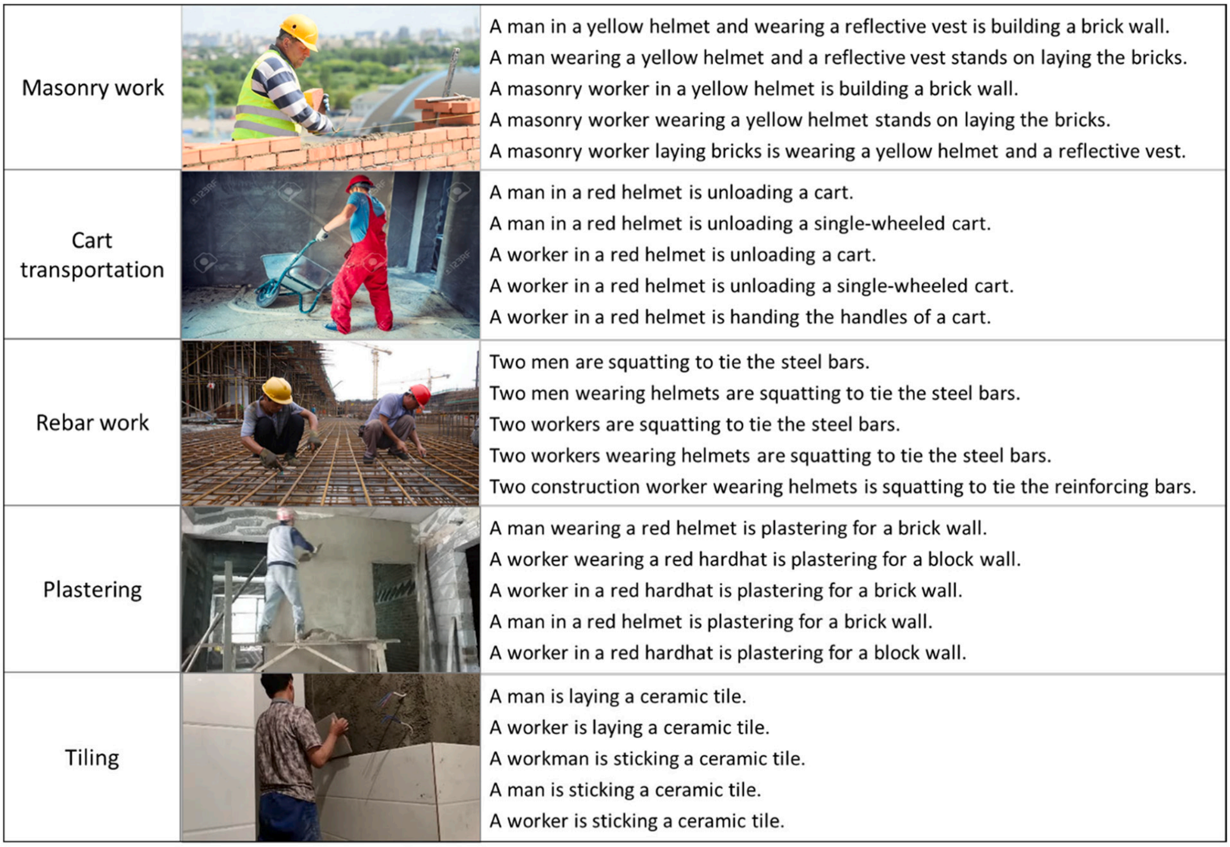


Fig. 5. Examples of the manual image descriptions.

Table 4
Overview of the created datasets.

	Activity categories	Image count	Sentence count	Word count
Dataset I	Cart transportation; Masonry work; Rebar work	480	2400	29,103
Dataset II	Cart transportation; Masonry work; Rebar work; Plastering; Tiling	6902	34,510	409,576

way, a basic theoretical framework for image captioning can be constructed by taking an image I as input and maximizing the likelihood $p(D|I)$ of producing a target sequence of words as a descriptive sentence D, where each word is obtained from a given dictionary to describe the image [27].

The main objective of modeling the image captioning is to maximize the probability $p(D|I)$ of the correct descriptive sentence from

$$\xi^* = \underset{\xi}{\operatorname{argmax}} \sum_{(I,D)} \log p(D \mid I; \xi) \tag{1}$$

where ξ represent the parameters of the model, I represents an image, and D is its correct descriptive sentence. Due to the unbounded length of descriptive sentence D, the joint probability is modeled over D_0, …, D_n by the chain rule [27,30], where n is the length of the sentence, a particular example being

$$\log p(D \mid I) = \sum_{t=0}^{n} \log p(D_t \mid I, D_0, \ldots, D_{t-1}) \tag{2}$$

where $p(D_t|I, D_0, \ldots, D_{t-1})$ represents one word generated at time t conditioned on a context vector, inferring the previous hidden state and the previously generated words. In the training process, the image-sentence (D, I) pairs are the training samples, with the sum of the log probabilities in Eq. (2) needing to be raised using stochastic gradient descent optimizers [40]. The loss during training is the sum of the negative log-likelihood of the correct word at each step, expressed as

$$L(I \mid D) = -\sum_{t=1}^{n} \log p_t(D_t) \tag{3}$$

this being minimized w.r.t. all the LSTM parameters, the CNN image embedder top layer, and word embedding W_e [32].

3.3.2. Model architecture

Fig. 8 depicts the general architecture of the image captioning model. A joint model system combining an “encoder” CNN with a “decoder” RNN is constructed. The “encoder” CNN is used for representing site images, by first pre-training it for an image classification task according to the scene elements and using the last hidden layer as an input to the RNN decoder that generates the sentences. Word embedding is one of the most popular representations of document vocabulary by capturing the context of a word in an original sentence bank, and its semantic and syntactic similarity in relation to other words.

1) *CNN, RNN, and Word embedding methods.* The VGG-16 [60] and ResNet-50 [61] is used as the basic CNN framework, as these perform well in visual recognition, with ResNet winning the 2015 ImageNet Large Scale Visual Recognition Challenge (ILSVRC) competition [61]. The performances of different DNN are compared in the experimental tests as they may be different with the same datasets.

The LSTM is used as the decoder RNN, as recent advances of image captioning have widely shown it to be the typical approach to model

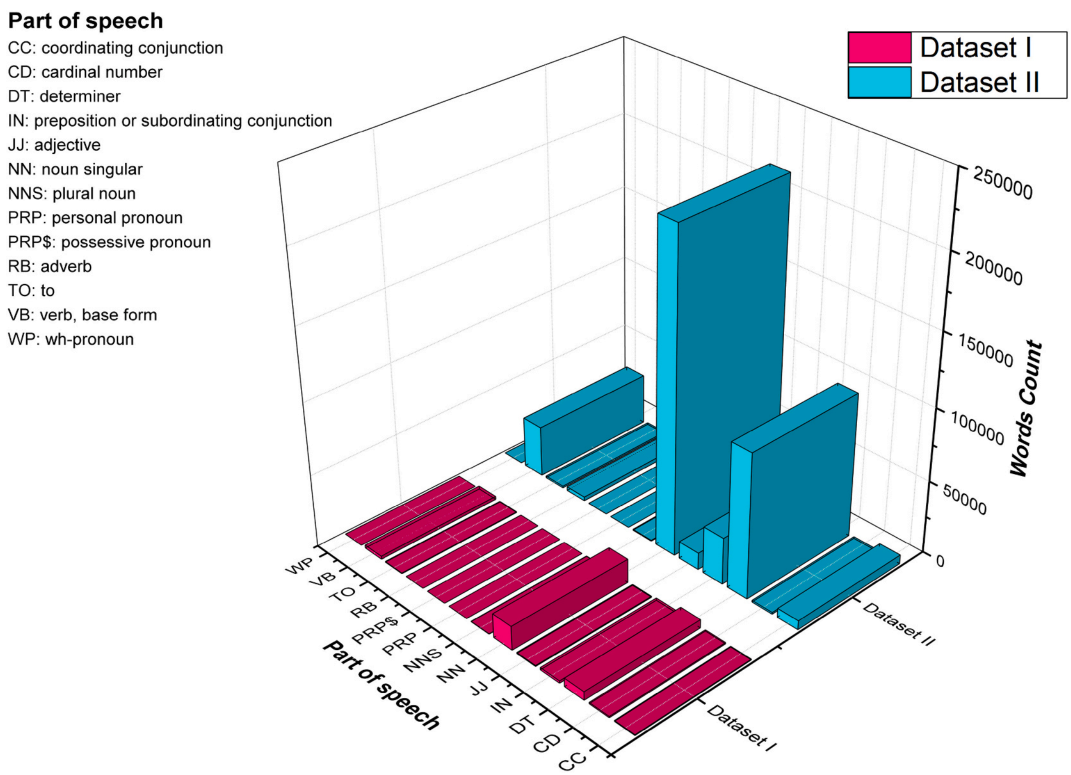


Fig. 6. Part-of-speech distribution in Datasets I and II.

time series vector $p(D_t|I, D_0, \ldots, D_{t-1})$ [27,30,55]. A common LSTM unit comprises a cell, input gate, output gate, and forget gate. The cell remembers values over arbitrary time intervals, and the three gates regulate the flow of information into and out of the cell (see [57] for more details).

The words are represented with a words-embedding model, which can capture a large number of precise syntactic and semantic word relationships [62]. Fig. 8 shows the specific embedding process used. A dense vector W_eD_t' is generated after the input word D_t' passes through two word-embedding layers W_e, whence the dense vector W_eD_t' continues to be passed on to RNN. Following [27,30], the dimension of each embedding layer is 512.

2) *Model learning strategies.* A joint model combining the CNN for image representation with the LSTM is trained to predict each word of a sentence (similar to [27,30]). The experimental tests to ensure the performance of these DNNs (i.e., CNN and LSTM) use batch normalization [63]. Beam Search is used as the inference method to generate a sentence in the model by following the recommendations of [27,30]. This is well-suited to scene labeling, because it explicitly accounts for the spatial extent of objects, conforms to inconsistency constraints from domain knowledge, and has a low computational cost [64]. The method iteratively considers the set of k best sentences up to time t as candidates to generate a $t + 1$ size of sentences and retain the resulting best k. Therefore, Beam Search can approximate $D = arg\,max_{D'}\, p(D'|I)$ well, and is used with beam size set as $k = 3$ (similar to [27,30]) in the experiments.

4. Experiments and results

Given the established model architecture with two DNN compositions (i.e., the VGG-16 "encoder" with LSTM "decoder", and ResNet-50 "encoder" with LSTM "decoder") and two created datasets (i.e., Dataset I and Dataset II), three experiments in this study are conducted, termed:

(1) *Experiment 1 (E#1)*: VGG-16 + LSTM @ Dataset I;
(2) *Experiment 2 (E#2)*: VGG-16 + LSTM @ Dataset II;
(3) *Experiment 3 (E#3)*: ResNet-50 + LSTM @ Dataset II.

At the current level of technology, the image captioning task is more challenging than the object classification and data-driven approaches [30]; training a deep learning model is still full of uncertainties. Since this is an initial trial of image captioning in the construction management context, this section focuses on describing the training process and evaluation results.

4.1. Setting for training

Two main tasks in training settings are to avoid overfitting and determine the hyperparameters.

The biggest challenge of these experiments is that the small dataset (compared with the computer-vision benchmark datasets) used for training the deep learning networks can result in overfitting. Several techniques were tried to deal with this. The conventional way (similar to [66]), called Fine-tuning, of initializing the weights of the CNN part of the model loaded from a pre-trained model (e.g., on ImageNet [65]) produced no obvious improvement in the evaluation scores. Then, similar to [27,30], the model-level techniques of Dropout [67] and model assembling were tried, as well as exploring the capacity of the model by trading off a few hidden units against depth. *Dropout* is used to simplify the structure of the networks automatically during the training process – the dropout degree relying on an initialized dropout rate, while *model assembling* is a one-off operation at the beginning of training by setting the initial number of layers or dimensions of the networks. As Fig. 9 indicates, Dropout and model assembling improve the evaluation considerably more the Fine-tuning in E#1.

Table 5 gives the necessary hyperparameter settings in the three experiments. Firstly, when experimenting on the tiny dataset in E#1, the structure of the networks of the model were actively simplified and the Dropout strategy adopted in the process of training to prevent

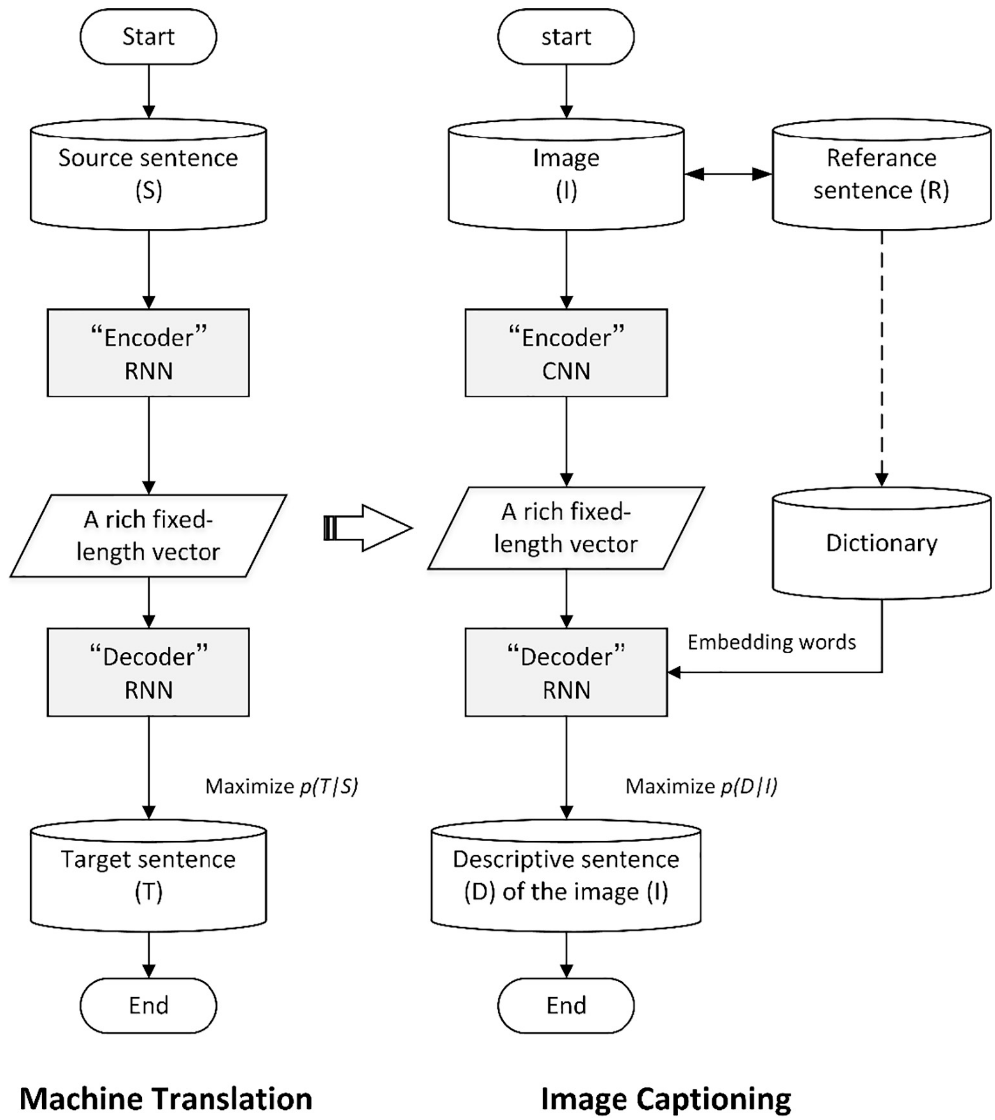


Fig. 7. Comparison of the machine translation and image captioning models.

overfitting. However, for the larger dataset in E#2 and E#3, only the appropriate network dropout rate was set in the training process. The dropout rate in E#3 is set lower than E#2 because the ResNet-50 has its skip module called "shortcut connections" [61], which can skip one or more layers during the training process and achieve better results. Secondly, the initial learning rate is a very important indicator for the initialization of model hyperparameters as it determines whether the loss function can converge and whether the correct rate tends to be roughly stable during training. Because of the small-size dataset in E#1, a larger learning rate is needed for the model loss function to converge more quickly. Also, batch normalization optimizes the deep learning training process [63], and the "Adam" optimizer was used to learn the network parameters [68]. Table 4 shows the number of iterations and training time in the three experiments. HD Graphics 630 (Kaby Lake GT2) and an Nvidia GeForce GTX 1080 GPU were used to train models on the small E#1 dataset and bigger E#2 and E#3 datasets respectively, implemented with activity recognition [24].

Figs. 10 and 11 show the accuracy and total loss in the training process corresponding to the best results of E#2 and E#3. A polynomial fit analysis (order = 8) [69] helps show the changing in accuracy and loss more clearly. This indicates that the fitting lines of training accuracy and loss stop early and tend to a uniform stability, the loss is convergent, and the value of the current accuracy in a stable state is close to the value of the subsequent test result. The generalization errors of the models in E#2 and E#3 are therefore small, indicating that the training process of the models could be effective [70].

4.2. Performance evaluation metrics

4.2.1. Automatic evaluation metrics for method validation

This section introduces a series of automatic evaluation metrics to measure the captioning efficiency in the experiments. For all metrics, higher values indicate a better configuration of the proposed method.

1) *Bi-lingual Evaluation Understudy (BLEU).* The machine translation community's BLEU [71] is a modified precision metric with a sentence-brevity penalty, calculated as a weighted geometric mean over different length n-grams. The common formulations of BLEU are BLEU-1, BLEU-2, BLEU-3, and BLEU-4, with BLEU-n meaning using 1-gram up to n-grams.
2) *Recall-Oriented Understudy for Gisting Evaluation (ROUGE).* The summarization community's ROUGE [72] is a package of measures for the automatic evaluation of text summaries using F-measures. As with BLEU, versions of ROUGE can also be defined by varying the n-gram count, expressed as ROUGE-N, and three other versions of ROUGE are ROUGE-L, ROUGE-S, and ROUGE-W, of which the

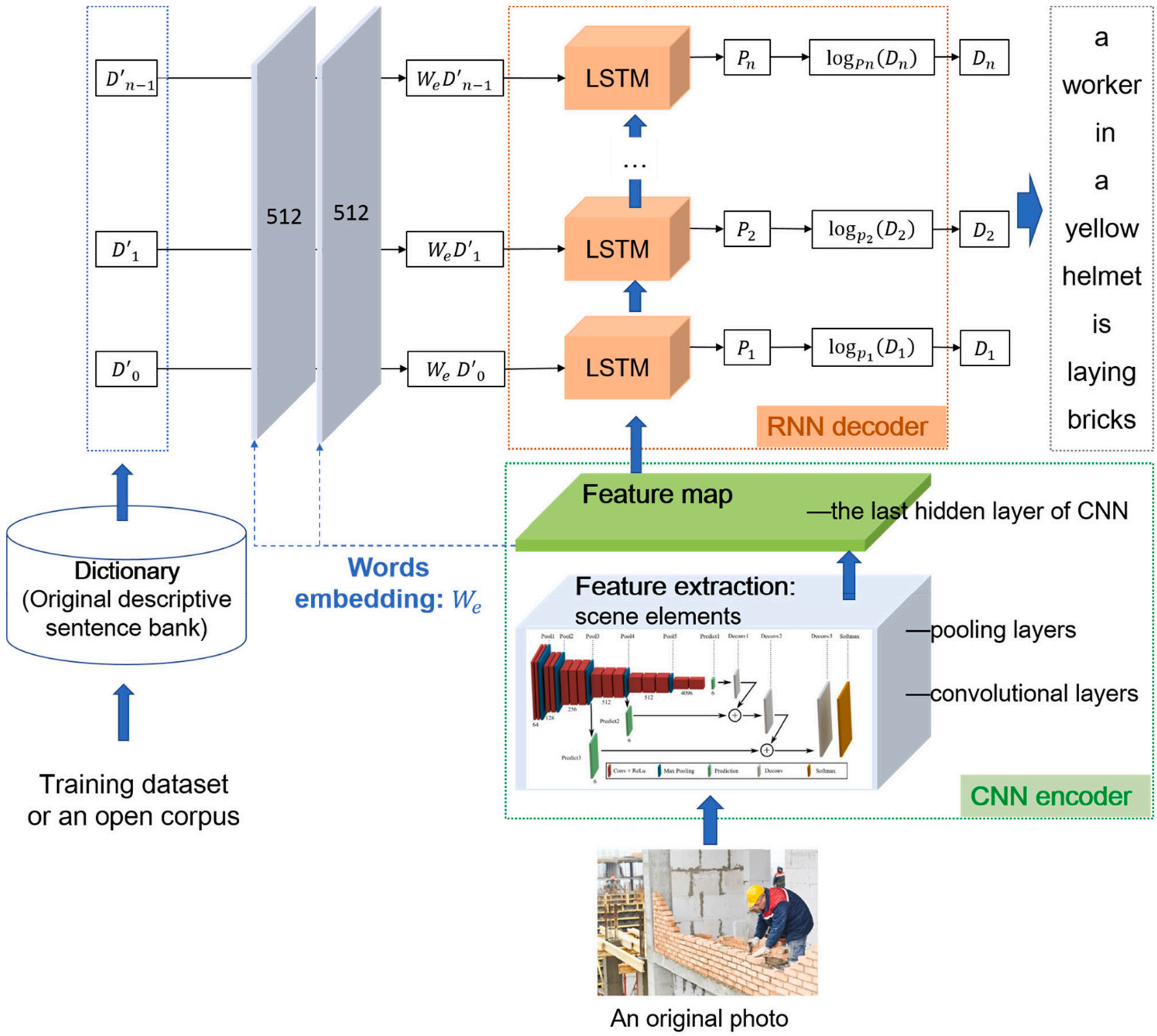


Fig. 8. General architecture of the image-captioning model.

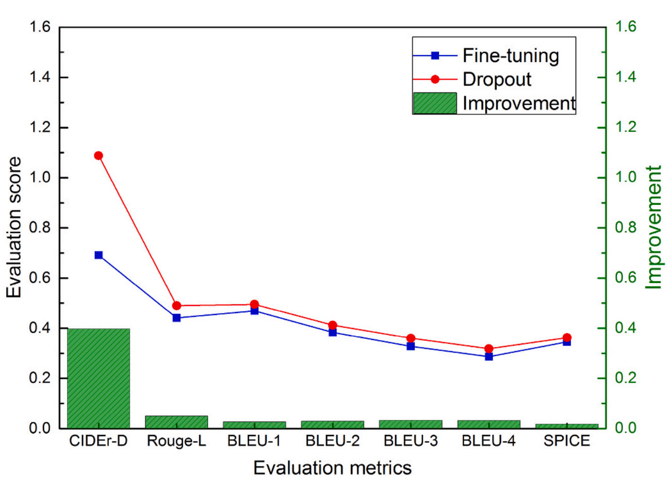


Fig. 9. Comparison of results of different overfitting avoiding strategies in E#1.

ROUGE-L is used in the present study.

3) *Consensus-based Image Description Evaluation (CIDEr).* CIDEr [73] captures the consensus by applying term frequency to inverse document frequency weights to n-grams in the candidate and reference sentences, which are then compared by summing their cosine similarity across n-grams. To help defend against the gaming of the CIDEr metric, CIDEr-D is proposed after improving the basic CIDEr metric.
4) *Semantic Propositional Image Caption Evaluation (SPICE).* Recently, a new automatic caption evaluation metric coined SPICE [74] is defined over scene images. Previous studies with extensive evaluations across a range of models and datasets indicate that SPICE captures human judgments over model-generated captions better than other automatic metrics.

4.2.2. Manual evaluation metrics for method testing

The performance of automatic captioning in the experiments can be reflected at the level of scene element. Drawing on the traditional information retrieval evaluation method [75], *Precision*, *Recall rate*, and *F-score* are common indicators to assess the correctness of generated captions at the level of scene word, given by.

$$Precision = \frac{TP}{TP + FP} \tag{4}$$

$$Recall = \frac{TP}{TP + FN} \tag{5}$$

Table 5
Some necessary hyperparameter settings for the three experiments.

Setting items		E#1	E#2	E#3
Model architecture	Encoder CNN	VGG-16	VGG-16	ResNet-50
	Decoder RNN	LSTM	LSTM	LSTM
	Dimension of the decoder layer	128	512	512
	Number of words embedding layers	2	2	2
	Dimension of each word embedding layer	512	512	512
	Initialization	NO	YES	YES
	Number of initialization layers	–	2	2
	Dimension of each initialization layer	–	512	512
	Beam search (size)	3	3	3
	Initial learning rate	0.1	0.001	0.001
	CNN drop rate	0.7	0.7	0.5
	RNN drop rate	0.5	0.5	0.3
Vocabulary dimensions		200	260	260
Optimizer		Adam	Adam	Adam
Batch normalization	Epochs	10	10	10
	Batch size	8	8	8
Training process	Training time	1 h 38 min 29 s	52 h 13 min 45 s	30 h 38 min 09 s
	Iterations	2200	28,360	28,360
	Training time(s)/iteration	2.68	6.63	3.89
Computer hardware		HD Graphics 630 (Kaby Lake GT2)	Nvidia GeForce GTX 1080 GPU	Nvidia GeForce GTX 1080 GPU

$$F - Score = (1 + \beta^2)\frac{Precision \cdot Recall}{\beta^2 \cdot Precision + Recall} \quad (6)$$

where the distributions of the scene elements retrieval results are as shown in Fig. 12. *Precision* represents the ratio of the amount of the scene element retrieved and relevant (*TP*) to the total amount of the scene element retrieved (*TP* + *FP*). *Recall rate* represents the ratio of the amount of the scene element retrieved and relevant (*TP*) to the total amount of the scene element relevant (*TP* + *FN*). For the *F-score*, the weight of the *Precision* and *Recall rate* can be controlled by adjusting β – the *Precision* rate is more important when $\beta < 1$, the *Recall rate* is more important when $\beta > 1$, and the *Precision* and *Recall rate* have equal importance (now called *F1*) when $\beta = 1$.

Given the possible retrieval deviation caused by the diversity of vocabulary application, there is a general assumption that the final evaluation score of each scene element is calculated based on combining the retrieval results of all the corresponding words; e.g., scene words such as "helmet" and "hardhat" are synonyms that correspond to the same object in images, so when calculating the performance score of this object, the two words are treated as the same.

4.3. Experimental results

The evaluation of the proposed image captioning method contains two parts. First, the effect of the pre-trained image captioning model is evaluated by using automatic evaluation metrics [71–74], and the efficiency of the image captioning method in the above three experiments is analyzed. Second, the testing results are evaluated by human analysis at the sentence and word level.

4.3.1. Validation results

Table 6 shows the best evaluation scores of each evaluation metric in the three experiments. The MS COCO caption evaluation results shown in Table-C5 (CIDEr-D of 1.196, Rouge-L of 0.573, BLEU-4 of 0.369, and SPICE of 0.215), which indicates the method's performance to be comparable with that of state-of-the-art computer vision methods in general (see http://cocodataset.org/#captions-leaderboard, referred to on 30 October 2019). The image captioning method has an evaluation score with CIDEr-D of 1.605, Rouge-L of 0.651, BLEU-4 of 0.491, and SPICE of 0.364, which is higher than the top score of the MS COCO caption evaluation results. Notably, the SPICE of 0.364 in E#1 is much higher than the other two experiments; this is because the number of scene element categories in Dataset I is smaller than Dataset II, and the

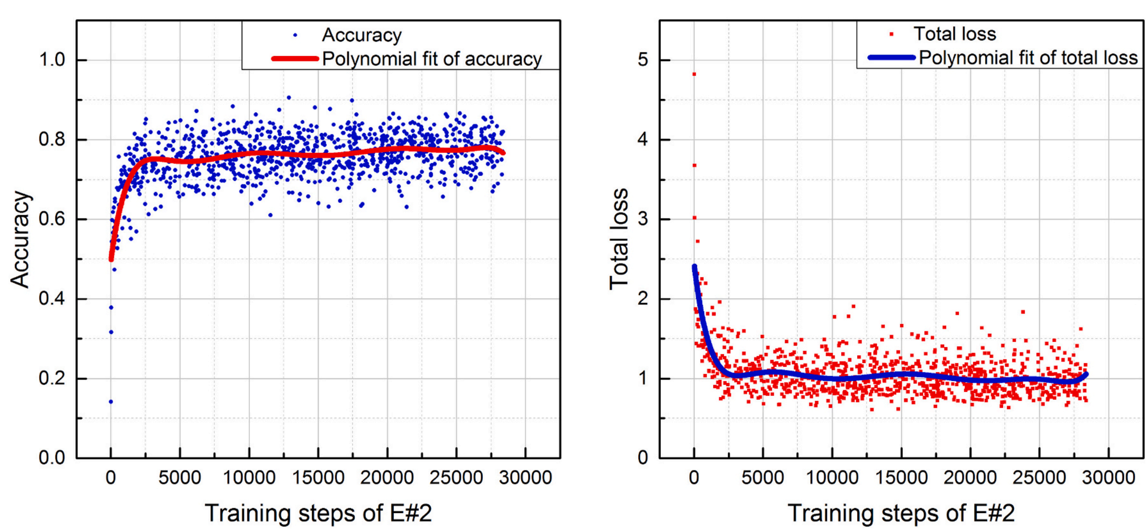


Fig. 10. Accuracy and total loss of the training process in E#2.

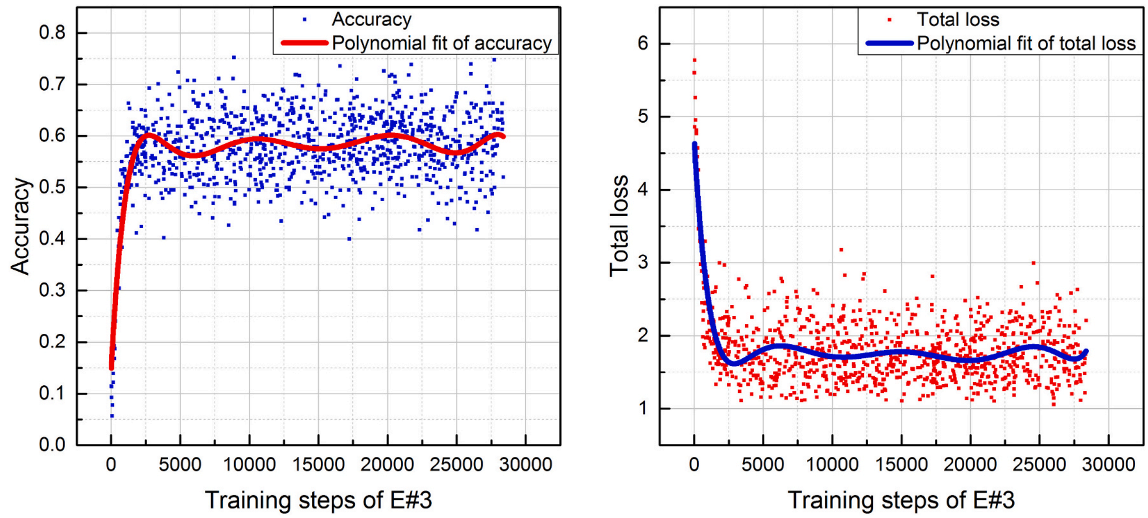


Fig. 11. Accuracy and total loss of the training process in E#3.

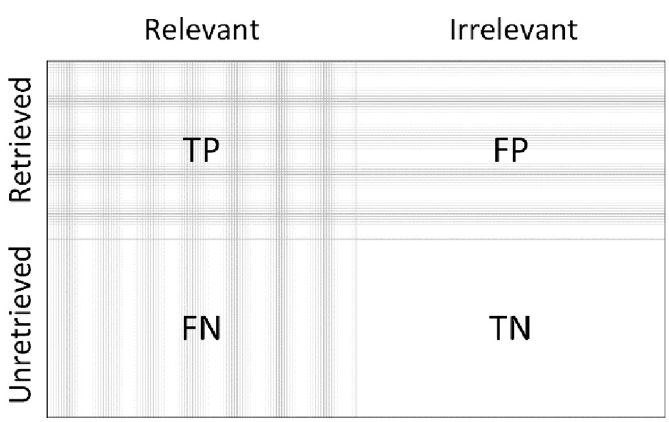


Fig. 12. Distribution of the scene element retrieval results.

Table 6
Highest values of each evaluation metric.

	Evaluation metric						
	CIDEr-D	Rouge-L	BLEU-1	BLEU-2	BLEU-3	BLEU-4	SPICE
E#1	1.088	0.501	0.518	0.427	0.36	0.318	0.364
E#2	1.545	0.651	0.658	0.576	0.522	0.482	0.349
E#3	1.605	0.616	0.680	0.595	0.537	0.491	0.354

probability of a single scene element in the Dataset I output to the sentence generation results is higher. These results are therefore comparable with the image captioning methods with benchmarking datasets in the CV community.

The E#1, E#2, and E#3 results are next compared and discussed to explore the influence of different datasets or different DNN compositions on the application of the image captioning method.

4.3.1.1. Results comparison between E#1 and E#2. Table 7 partially presents the valid results in E#1 and E#2, with validity meaning the achieved generalization ability of the model confirmed by the convergence of the total loss, accuracy stability, and accuracy comparison results with no significant difference between the process in training and testing. The underlined scores are the best results of the corresponding evaluation metrics. The best results do not necessarily appear in the same configuration of experiments because the evaluation dimension of different metrics is different, as previously introduced in Section 4.2.1.

In Table 7, bolded results are the average scores of all valid results of each experiment. Fig. 13 intuitively shows the differences between E#1 and E#2 by comparing the average value, and the D-value represents the difference in evaluation scores between E#1 and E#2, which used the same DNN compositions with two different datasets. It is found that the effect of the same model on Dataset II (in E#2) is significantly better than that on Dataset I (in E#1), because Dataset II has a larger size of image-sentence samples and richer categories of scene elements than Dataset I. The results clearly accord with the convention of the learning of a deep learning model in the computer science field. The results also indicate that the constructed method relies heavily on datasets.

4.3.1.2. Comparison of E#2 and E#3 results. The evaluation results of the same dataset in E#2 and E#3 were compared to explore the difference in performance of the two DNN compositions (i.e., VGG-16 with LSTM, and ResNet-50 with LSTM). The experimental results in Table 8 and comparison results in Fig. 14 indicate that the E#3 results are slightly better than E#2. However, the difference between the two experiments is not significant. The Rouge-L result in E#2 is better than that in E#3, which generally accords with the results of previous research [76]. The difference between VGG-16 and ResNet-50 in gaining robustness against the label noise of caption datasets is that ResNet-50 performed better than VGG-16 [76]. The difference in application is that the CNNs within the joint deep learning model are used for representing the vision of images.

In conclusion, the evaluation results of the proposed image captioning method are comparable with the CV community's state-of-the-art results. The grouping comparison results show that the difference in the influence of the two datasets on the method is significant: the dataset with a larger size and richer categories of samples performed better. Moreover, the difference in influence of the two DNN compositions on the method is not significant, although the E#3 results are generally slightly better than E#2. It is shown that the proposed method relies on the dataset more heavily than the DNN compositions of the model. The definite difference in performance of sentence generation between E#2 and E#3 not only depends on the quality of the datasets and the established model architecture itself but is also related to the research conditions and other uncertainties that need to be further explored in follow-up studies.

4.3.2. Testing results

4.3.2.1. Human evaluation. Keeping the training setting in E#1, E#2,

Table 7
Experimental results corresponding to the configurations in E#1 and E#2.

Configuration		Evaluation metric						
		CIDEr-D	Rouge-L	BLEU-1	BLEU-2	BLEU-3	BLEU-4	SPICE
E#1	1	0.933	0.497	0.479	0.388	0.329	0.287	0.348
	2	1.088	0.490	0.495	0.412	0.36	0.318	0.283
	3	0.950	0.501	0.484	0.395	0.336	0.293	0.299
	4	0.972	0.495	0.518	0.427	0.355	0.296	0.297
	5	0.868	0.482	0.468	0.375	0.316	0.273	0.364
	6	0.898	0.497	0.506	0.391	0.321	0.269	0.347
	7	0.843	0.451	0.446	0.345	0.288	0.25	0.314
	8	0.815	0.438	0.464	0.355	0.289	0.247	0.348
	9	0.934	0.499	0.48	0.391	0.332	0.288	0.349
	10	0.893	0.493	0.475	0.382	0.323	0.279	0.345
	11	0.796	0.423	0.438	0.336	0.277	0.234	0.318
	12	0.611	0.389	0.419	0.311	0.245	0.194	0.276
	13	0.657	0.43	0.449	0.362	0.3	0.253	0.272
	14	0.631	0.426	0.443	0.355	0.292	0.246	0.269
	15	0.972	0.495	0.518	0.427	0.355	0.296	0.297
	Avg.	**0.857**	**0.467**	**0.472**	**0.377**	**0.315**	**0.268**	**0.315**
E#2	1	1.456	0.64	0.621	0.511	0.424	0.357	0.341
	2	1.493	0.641	0.614	0.502	0.420	0.357	0.344
	3	1.502	0.639	0.615	0.502	0.420	0.357	0.344
	4	1.493	0.641	0.614	0.502	0.420	0.357	0.344
	5	1.496	0.648	0.617	0.514	0.435	0.369	0.326
	6	1.467	0.561	0.671	0.538	0.438	0.366	0.272
	7	1.497	0.645	0.620	0.509	0.424	0.361	0.342
	8	1.130	0.548	0.545	0.428	0.337	0.267	0.237
	9	1.545	0.651	0.658	0.576	0.522	0.482	0.318
	10	1.508	0.642	0.619	0.511	0.423	0.362	0.349
	Avg.	1.459	0.626	0.619	0.509	0.426	0.364	0.322

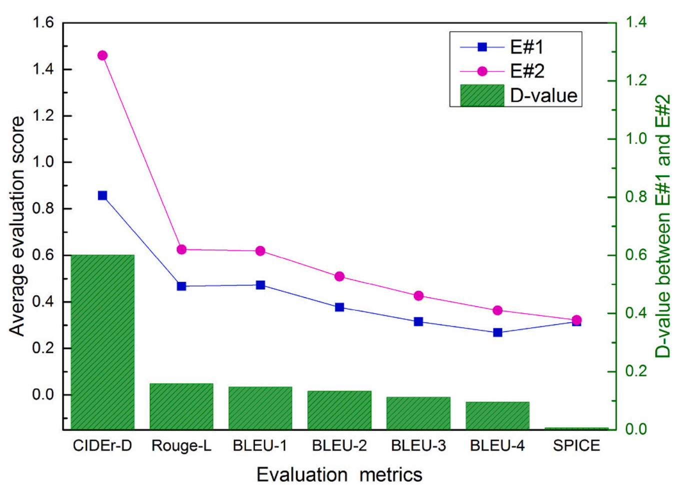


Fig. 13. Comparison of E#1 and E#2 average-score results.

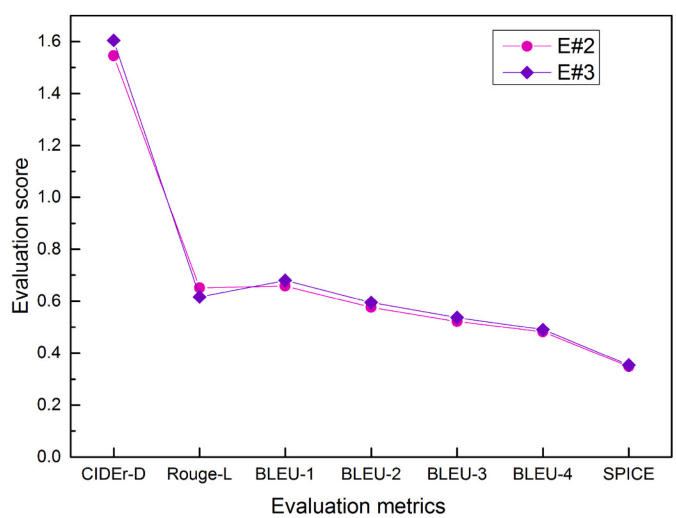


Fig. 14. Comparison of E#2 and E#3 highest scores.

and E#3, the images in the testing sets from actual construction sites were annotated automatically. Table 9 shows some examples of the testing results generated by the Beam Search decoder [64]; good examples marked with a G achieved the top evaluation scores, and poor examples with some obvious errors and incompleteness are marked with B1 and B2. Some sentences (underlined) containing both good and bad samples in Table 9 are not presented in the training set. This caption-generating model can deal with diverse situations, indicating its generalizability. Fig. 15 gives an intuitive comparative analysis of the image captioning results, grouped by human rating. In the first row, the sentences describing the scene pictures with the

Table 8
Experimental results with the highest metric score corresponding to the configurations in E#2 and E#3.

Configuration		Evaluation metric						
		CIDEr-D	Rouge-L	BLEU-1	BLEU-2	BLEU-3	BLEU-4	SPICE
E#2	1	1.545	0.651	0.658	0.576	0.522	0.482	0.318
	2	1.508	0.642	0.619	0.511	0.423	0.362	0.349
E#3	1	1.605	0.616	0.680	0.595	0.537	0.491	0.354

Table 9
Example sentences from the caption generation results.

Example sentence	Type[a]	Source
Two men are tying the steel bars.	G	E#1
A man wearing a yellow helmet and a reflective vest	G	E#1
A man is squatting to tie the steel bars.	G	E#1
A man in a yellow helmet is squatting to tie the steel bars.	G	E#1
A man in a yellow helmet is pushing a two-wheeled cart.	G	E#1
A man wearing a yellow helmet and a reflective vest	B2	E#1
A masonry worker wearing a yellow helmet and a reflective vest is pulling a two-wheeled cart.	B1	E#1
A man is fixing a brick.	G	E#2
A man wearing a yellow helmet is fixing a brick.	G	E#2
A man wearing a red helmet is cleaning side mortar of the brick wall.	G	E#2
A man wearing a yellow helmet and an orange reflective vest is fixing a brick.	G	E#2
A man wearing a yellow helmet and a yellow reflective vest is fixing the ceramic tiles.	G	E#2
A worker wearing a red hardhat is cleaning side mortar of the brick wall.	G	E#3
A worker wearing a yellow hardhat is cleaning side mortar of the brick wall.	G	E#3
A worker wearing a yellow helmet is fixing a brick wall.	G	E#3
A worker wearing a yellow hardhat is fixing a brick wall.	G	E#3
A man wearing a yellow helmet and a yellow reflective vest is fixing the ceramic tile.	G	E#3
A worker wearing a yellow helmet is fixing a yellow helmet is cleaning side mortar of the brick.	B1	E#3
A worker wearing a worker wearing a yellow helmet is cleaning side mortar of the brick.	B1	E#3
A worker wearing a brick	B2	E#3
A worker wearing a brick wall	B2	E#3

[a] G denotes a good example of the top five in each experiment, B1 denotes a poor example with logic or grammar errors, and B2 denotes a poor example with an incomplete description.

completeness of a scene have no errors. In other rows, all the sentences generated have mismatching errors with the visual information in presented pictures, of which the corresponding error words are marked in red boxes.

4.3.2.2. Correctness at the scene element level. Eighty pictures (containing three categories of construction activity of cart transportation, rebar work, and masonry work) from actual construction sites as the testing sets are automatically annotated by the learned image captioning model in E#1. According to the classification of the scene elements in Fig. 4, the correctness in *Precision*, *Recall rate*, and *F-Score* of each concerned scene element and each category are calculated. A similar previous study calculated the *F*-scores by semantic proposition subcategory (object, relation, attribute, color, count, size, etc.) [77]. The results are presented in Table 10; the average values of the *Precision*, *Recall rate*, and *F-score* in the same category of scene elements are also presented. For comparison, the column chart in Fig. 16 presents a comparison of retrieval correctness between different categories of scene elements, showing that the retrieval correctness of *Object I*, *Object III*, and *Attributes* are better than the other two categories. The most likely reason is that the proposed model architecture for caption generation is based on the encoder-decoder framework with two DNNs, which corresponds to the statistical approach for NLG [40]. The final prediction correctness of objects is closely related to the adequacy of the corresponding training samples, which accords with statistical law of large numbers, where the larger the basic sample size is, the more accurate the learning results will be. As Fig. 16 shows, the correctness of different categories of scene elements on *Precision*, *Recall rate*, and *F-score* is consistent with their different frequency in the original data (pictures and sentences), which is represented as the blue line chart in Fig. 16.

In summary, the experimental tests indicate the following results. The automated evaluation results show the method's performance to be comparable with that of state-of-the-art computer vision methods in general. The group comparison results indicated that the method relies on the dataset more heavily than DNN compositions of the model (see Figs. 13 and 14); the dataset with a larger size and richer categories of samples performed better. The tests by human evaluation proved the generalizability of the method. Although the automated image captioning is more efficient in time than manual annotation, there is still some scope for improvement from a practical perspective as there are errors in the generated sentences, as shown in Table 9 and Fig. 15. Table 10 and Fig. 16 also show that correctness varies between different categories of scene elements. From the perspective of statistics, this may be due to the limitations of the original image captioning datasets in that: 1) the number of each scene element is small; 2) there are not enough scene categories, and the number of categories is quite varied.

5. Discussion

5.1. Knowledge contribution

This study makes significant contributions to current scholarly literature. First, the sparse existing literature considers the automatic manifestation of the construction activity scene as an integral whole. This study represents one of the first attempts to bridge this important void by employing recent advances of image captioning, rooted in the fields of CV and NLG, to manifest the scene as an integral whole, with both visual information and natural language sentences. The method adopts a joint deep learning model by integrating a CNN "encoder" and an RNN "decoder" and other related function modules (e.g., word embedding model, and optimizer). Two dedicated image captioning datasets are created for training the proposed model and validating the method, which imposes such major challenges as overfitting problems during the model learning process and operation testing. These are overcome and solutions are presented containing the hyperparameter settings. The experiment results validated the method by showing its performance to be comparable with that of state-of-the-art computer vision methods in general.

Second, a novel linguistic description schema of construction activity scenes is pioneered, which guided the dataset production in the follow-up experimental study. The originality and easy applicability of the proposed schema is shown as follows. 1) It bridges the gap between the image captioning method and scenario applications at the level of information formats by structurally linking the visual information of scene elements with the corresponding natural language words or phrases. It is also generic and can be applied to other tasks that combine the techniques in the fields of CV and NLP (e.g., text-based image retrieval [3,78]). 2) The three steps for manifesting the construction scenes with descriptive sentences, including deconstructing the scene image, matching words with the scene elements, and forming a

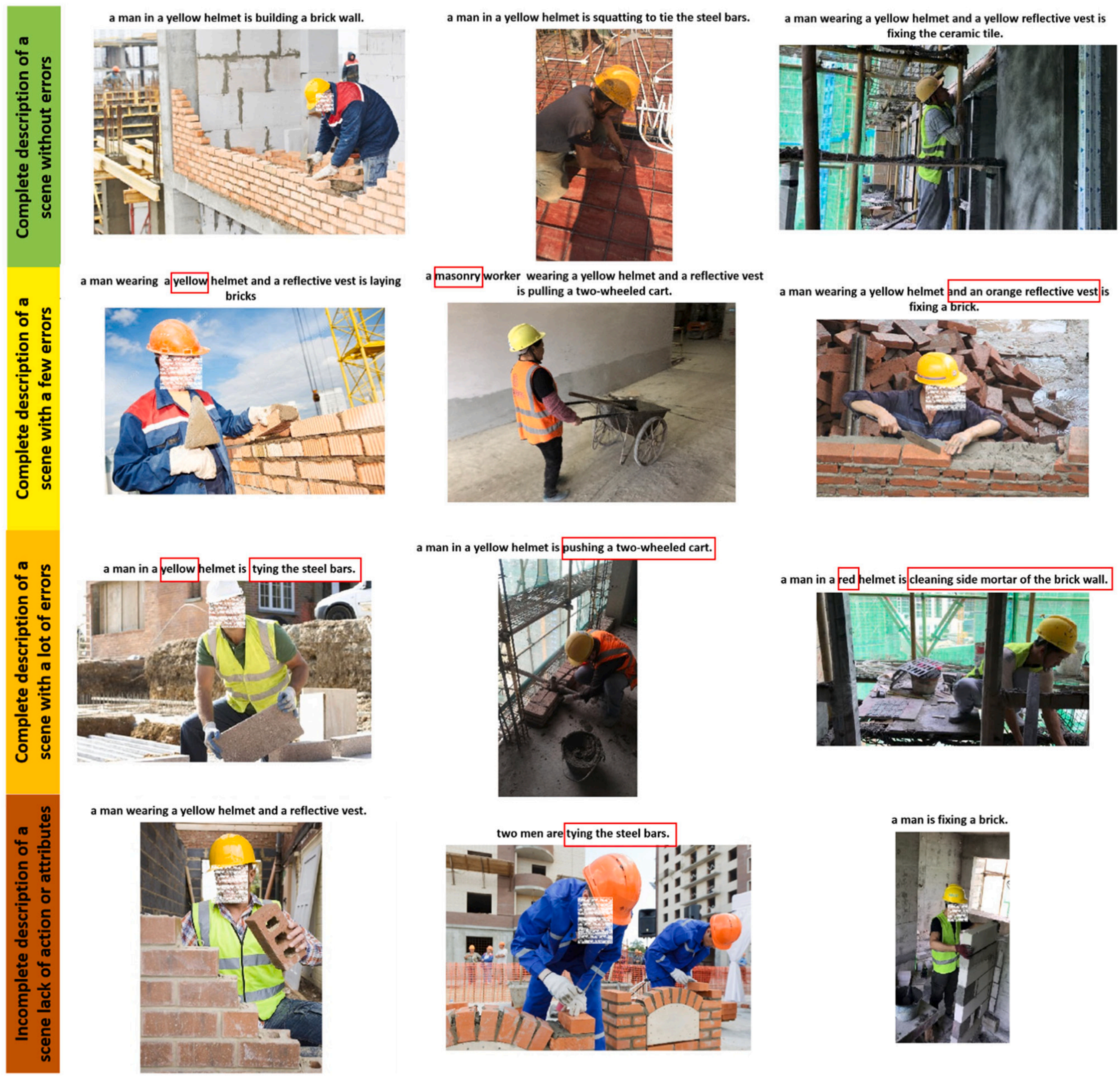


Fig. 15. Intuitive comparative analysis of captioning results.

Table 10
Distinguishing the *Precision*, *Recall*, and *F-Score* between different scene elements.

Categories	Scene elements	*TP + FP*	*TP + FN*	*TP*	*Precision*	*Recall*	*F1*	Average/category		
								Precision	*Recall*	*F1*
Object I	Worker/man	80	79	79	0.988	1.000	0.994	0.988	1.000	0.994
Object II	Cart	15	45	5	0.111	0.333	0.167	0.273	0.203	0.186
	Brick	33	16	6	0.375	0.182	0.245			
	Bars	32	9	3	0.333	0.094	0.146			
Object III	Helmet	73	79	73	0.924	1.000	0.961	0.631	0.843	0.707
	Reflection vest	35	71	24	0.338	0.686	0.453			
Relationships	Laying/build	31	14	5	0.357	0.161	0.222	0.288	0.210	0.190
	Tie	32	8	3	0.375	0.094	0.150			
	Pull/push	16	45	6	0.133	0.375	0.197			
Attributes	Color – “yellow”	36	72	32	0.444	0.889	0.593	0.564	0.902	0.688
	Count of workers – “1”	59	79	54	0.684	0.915	0.783			

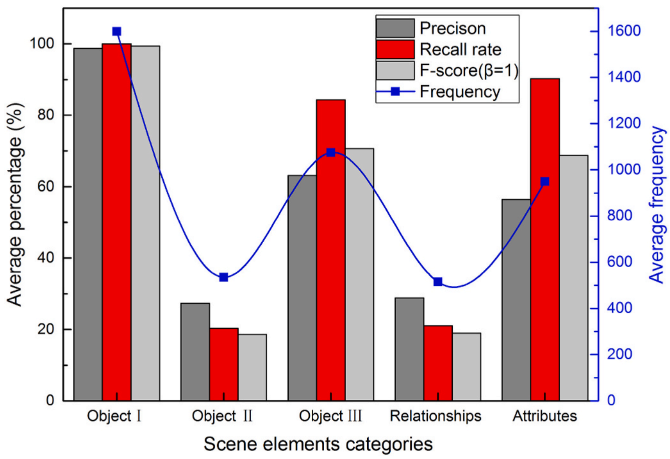


Fig. 16. Correctness comparison of different categories of scene elements.

descriptive sentence, are easily implemented either manually or by using NLP tools (e.g., NLTK toolkit [59] in Fig. 6).

5.2. Practical implications and application challenges

The practical implication and application challenges of the proposed image captioning method are discussed in this section.

1) *Practical implications.* The proposed method addresses the two low-efficiency problems of image application. First, it can manifest the activity scene of a site image with a complete sentence. Managers can obtain objective information of the construction activity state by scanning the generated image captions centrally. For example, a worker's safety state can be estimated by analyzing the distribution of such scene words as “helmet”, “hardhat”, or “reflective vest”. Second, the lack of content indexing method for managing random images leads to the low reuse rate of these images. The captions of images generated by the proposed method can be used as indices of images, which can assist in managing random site images by image classification and image retrieval based on scene words in captions. By repeated extraction and utilization of visual information from images for different image-based management objectives, the value of site images can be improved considerably.
2) *Application challenges.* Given the proposed method is based on a supervised learning-based model with multiple DNNs and manual datasets for training, two application challenges are apparent. First, the accuracy needs to be improved. The application performance depends on the quantity and quality of the training dataset and the inherent performance of the model architecture. The still-developing technologies of image captioning and difficulties in the production of big datasets in the context of the construction industry may be inhibiting accuracy, especially compared to the usual simple object detection and image classification tasks developed by the CV community. Second, the difficulty of cross-scenario applications of the method needs to be addressed. Natural language sentences in the image captioning dataset of different scenarios need to be diversified and generalized. Even for the complex world of construction site activities, the proposed method is based on limited categories of construction activity, as well as the limited number and diversity of words. Developing a more comprehensive method may require entirely new lexicons, continuously updated with the additional polysemy and synonymy words, which will constitute a significant application challenge.

5.3. Research limitations

This study is limited to three categories of site activities in Dataset I and five in Dataset II, and the proposed method is fit for application in the scenarios involving these activities. Moreover, while two dedicated datasets created in this study can serve for both practice and further research, making the proposed method more practical in the context of construction management means that they need to be constantly supplemented to expand their sample size and categories. Ideally, this would be an industry-based process, but is unlikely at present because of the expense involved in creating such a dataset for general objects and services – the lack of well-known datasets for testing optional approaches already being a chronic problem [24].

The study is also limited to two DNN compositions (i.e., VGG-16 with LSTM, and ResNet-50 with LSTM) of the image captioning model, whose difference in results is not significant. There are more uncertainties in the training process of a joint model with multiple DNNs, such as a many hyperparameter settings and the users' degree of experience in making adjustments. Therefore, future studies could further explore the difference between different DNN compositions of this image captioning model.

6. Conclusions

This paper introduces an automated method for manifesting construction activity scenes via image captioning employing recent advances of image captioning in the fields of CV and NLG. A joint image captioning model is established by integrating a CNN (i.e., VGG-16 [60] or ResNet-50 [61]) “encoder” for image representation with the Skip-gram model [62] for word embedding, and an RNN (i.e., LSTM [57]) “decoder” for sentence generation. The testing results by human evaluations demonstrate the feasibility of the proposed method's application in practice at the current technical level, although there is still scope for improvement. The discussion of practical implications and application challenges also indicates the need to constantly improve the method by extending the dedicated datasets and improving the performance of the model structure.

Declaration of competing interest

We have no financial and personal relationships with other people or organizations that could inappropriately influence (bias) the work or state, which is reported in this paper for publication consideration by Automation in Construction.

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Appendix A. Supplementary data

The pre-trained models and datasets are publicly available at https://github.com/HannahHuanLIU/AEC-image-captioning.

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